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GEOMETRY OF REASONING TRAJECTORIES IN RECURRENT-DEPTH TRANSFORMERS: LATENT DYNAMICS IN HUGINN-3.5B

Abstract. We present an inference-only analysis of latent trajectories in the Huginn-3.5B recurrent-depth transformer [1]. Without fine-tuning, hidden states are extracted at each recurrence step via hooks on core_block[-1] and reduced to geometric, dynamical and topological descriptors testing three hypotheses. Answer-token trajectories predominantly converge exponentially to fixed points, consistent with Huginn's design and with independent measurement [2]; no depth-dependent winding is found, refuting the naive operationalisation of H2. We identify steps-to-settle as a proxy for effective computation that tracks state retention rather than prompt length, confirmed by a length-matched control whose correlation reverses sign with task type (+0.558 against −0.576). Two-scale analysis [3] shows acceleration correlating with depth ($\rho = 0.997$) and outperforming winding, while reproduction on real FineWeb text gives oscillatory rather than spiral dynamics and shows that orthogonality is not achieved without step normalisation. Measuring the same trajectories with a projection-free rotation statistic adds two results: the conspicuous period-6 structure is selected by one instruction word at fixed token count and is causally controlled by a single embedding row; and much of what we scored as capability was answer formatting, first-token scoring reading 0.125 where the answer is present in 0.781 of generations. The lines meet in a timing fact: the answer is fixed some twelve unrolls before the regime becomes measurable.

§1. Introduction

Test-time scaling has revived interest in recurrent-depth transformers [1]. Unlike chain-of-thought reasoning in token space, Huginn refines a state in $\mathbb{R}^{5280}$, applying a shared block r times, $\mathbf{h}_{t+1} = \mathbf{R}(\mathbf{h}_t; \mathbf{e})$, with

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$\mathbf{e}$ a fixed context encoding; empirically trajectories settle, loop or drift. The architecture initialises $\mathbf{h}_0$ randomly, which "promotes convergence to a steady state independent of initialization, i.e., path independence" [1]. We ask whether the geometry of these trajectories encodes computational depth and — separately — whether the quantities we score as capability measure what we take them to. Three hypotheses are tested.

H1 (regimes): each token's latent path falls into one of several regimes — settling, looping or drifting — distinguishable by recurrence and topological signatures. H2 (loops encode depth): when the path loops, its winding number grows with the reasoning steps the task requires. H3 (contraction prevents state tracking): if the update is strictly contractive the model cannot maintain a running count, so stateful tasks require looping or drifting.

The results divide into three lines, the third reinterpreting the first two. We contribute a length-matched control showing effective computation grows with state retention rather than prompt length; the finding that the conspicuous geometric structure is selected by one instruction word at fixed token count and causally controlled by one embedding row, with a validated scalar instrument for it; and a decomposition of measured capability into answer availability and answer format that reproduces the published benchmark number under its authors' protocol, together with a timing result placing regime and answer in non-overlapping windows.

What this report does not offer, stated once: every number comes from one model, one checkpoint and one seeded state per condition, so nothing here generalises to recurrent-depth models at large; several nulls rest on 12–24 paired units, excluding large effects and not small ones; and the behavioural test of H3 by state injection is absent rather than negative. Section 9 is explicit about the rest.

§2. Model, Protocol and Metrics

All experiments use Huginn-3.5B inference-only (revision bb6621b6. . . , transformers 4.53.3): a two-layer prelude runs once, a four-layer core block iterates r times, then a coda, a final normalisation and the head decode, with $\mathbf{e}$ re-injected at every unroll. Following the DEQ reading [6] we treat $\mathbf{e}$ as the map's parameter and $\mathbf{h}$ as its state, the block index never entering the block computation. We hook core_block[-1] at each unroll and seed every forward, the initial state being drawn from an unseeded generator by default.

Metrics. Winding number, accumulated angle in a per-trajectory 2D-PCA projection over 2π; steps-to-settle, the first step with $\|\Delta h_t\| < 0.1$, our proxy for effective computation; shape (settle/loop/drift); acceleration $a(k) = \|\Delta(k) - \Delta(k-1)\|_2$ and step orthogonality $\cos\theta_k$, the cosine between successive steps $\Delta(k)$ and $\Delta(k-1)$ [3]; and persistent homology H_1 via ripser [12]. The Lyapunov exponent and spectral radius $\rho(J)$ are not computed; convergence proxies stand in, a limitation for H3. Note ρ denotes Spearman correlation throughout except in $\rho(J)$.

Rotation power. Winding is read off a projection, so we add a projection-free measure to cross-check it. Taking the discrete Fourier transform along the unroll axis in the full 5280-dimensional state, rotation power P_6 is the fraction of non-DC power in the period-6 bin over a 24-unroll tail — one scalar, no basis choice. It reproduces its own spectral definition to 4×10^{-8}; one threshold $P_6 > 0.6677$ reproduces the hand-assigned rotate/settle label on 691 of 696 trajectories and 688 of 696 held out; and rotating prompts peak at period 6 specifically. Two cycles are needed, so 12 unrolls is the shortest window on which P_6 is defined — a bound that matters in §6.

Datasets. PARARULE-Plus (depths 2–5, 10 per depth); a running ±1 sum (lengths 2–32, 5–15 each); parity (4–48, 10 each); running-max (4–48, 8 each); 1000 FineWeb texts; and the regime sweeps below (16 nouns, 384 orbits, token count fixed at 53).

§3. Depth, Looping and Effective Computation

3.1. E1: Winding does not track reasoning depth. All 40 PARARULE-Plus answer-token trajectories (depths 2–5, $n = 10$ per depth) settle. Winding against depth gives $\rho = +0.20$ per level (n.s.) and -0.037 per row ($p = 0.82$); partialling by length, -0.234 ($p = 0.15$). Steps-to-settle grows weakly, $20.2 \to 22.0$. Multi-step symbolic logic does not induce looping.

3.2. E2: Counting, and a length confound. On a running ±1 sum, $n_{\text{ops}} \in \{2\dots32\}$, 5 seeds, all trajectories settle but steps-to-settle grows with n_{ops} ($\rho = 0.718$, $p < 1e{-}5$). Winding also correlates ($\rho = 0.554$, $p = 0.002$) — yet partialling by length collapses it to $\rho = 0.04$, $p = 0.82$. The winding correlation is driven by prompt length, which is what makes the next experiment necessary.

3.3. E3: A length-matched control. Track and local conditions share identical body text and differ only in the question (“What is the sum?” against

"What was the last instruction?"), matching length by construction. With 15 seeds:

Table 1. Length-matched control (15 seeds): sign is set by task type, not length.

Condition	Metric	ρ	p
Track (requires accumulation)	steps $\sim n_{\text{ops}}$	$+0.558$	$< 1e{-8}$
	winding $\sim n_{\text{ops}}$	$+0.324$	0.002
Local (last step only)	steps $\sim n_{\text{ops}}$	-0.576	$< 1e{-8}$
	winding $\sim n_{\text{ops}}$	$+0.054$	0.61 (n.s.)

At identical length the sign of the correlation is set solely by task type — state retention increases effective computation, pure retrieval decreases it — the strongest evidence here for H3's computational half.

3.4. E4: Loops are a budget symptom. With a reduced budget (num_steps = 16) loops appear — 7 of 8 at $n_{\text{ops}} = 24$, 5 of 8 at 48 — while at num_steps ≥ 24 all settle. Looping is a symptom of insufficient budget, not of task depth.

3.5. E5: The signature generalises to parity. On parity, $n_{\text{ops}} \in \{4 \ldots 48\}$, 10 seeds: winding is null ($\rho = 0.162$, $p = 0.22$) while steps-to-settle again tracks length ($\rho = 0.615$, $p < 1e{-6}$). The signature is not specific to arithmetic.

§4. Instruction Form Selects the Regime

4.1. E6: One instruction word switches the map. Holding the sequence, required answer, marker and token count fixed at 53, only the instruction noun varies. "Report the largest symbol" rotates 36/36 paired cells; "largest element" and "largest item" rotate 0/36. The selector is not semantic (neighbours of symbol rotate 0/144, of element 96/120), not orthographic (form-matched symptom, cymbal, symmetry rotate 72/72), and not tokenisation (the 53-token group splits exactly 50%). Editing a single embedding row, with the prompt string and every token identifier untouched, flips the regime, both gates bit-exact at $0.000e{+}00$. Swapping $\mathbf{e}$ mid-trajectory, 62/64 switches take at points from unroll 1 to 48 — the regime follows the parameter the map currently has, not its history — entering rotation taking a median 15.5 unrolls against 1.0 to leave.

Scope. The within-regime control failed and norm-matching rescued only one noun pair of two, so the licensed statement is that the row determines the regime within a small, direction-dependent neighbourhood.

4.2. E7: The regime belongs to the prompt, not the position measured. E1–E6 read one token position while the published figures are drawn at interior ones, so we re-measured P_6 at every position: rotating prompts rotate at 62.0% of theirs (0.614–0.623, $n = 9$), settling prompts at 0.000, no overlap. Onset is a single boundary — position 20 of 53 in symbol, where the second rule clause begins, not the noun at 14 — so the shared answer-token restriction does not bias the regime labels.

§5. What the Accuracy Measurements Measured

5.1. E8: Format, not availability. Scoring the first generated token reads 0.125 where the answer is actually present in the generated text in 0.781 of cases, against a measured false-positive rate of 0.042–0.152. What displaces it is prose: The opens 39.4% of 1260 generations. Depth worsens the metric while the output improves — across $r = 2, 4, 8, 16, 32$ the rate of opening with The runs $0.032 \to 0.099 \to 0.631 \to 0.560 \to 0.647$ ($p = 1.09e{-}12$) while opening with the gold does not move ($p = 0.515$). Literally, sub1 with gold 3 produces "The answer is 4 - 1 = 3", scored wrong.

Supplying the format recovers the published benchmark number: the ARC-Easy figure of 0.699 at $r = 32$ is produced zero-shot, normalised, with no option list, and scored that way we obtain 0.658; five in-context examples under option-letter argmax independently give 0.723 ($p = 3.35e{-}07$). Instruction fails where demonstration works — "reply with only the answer" gives no improvement, \boxed{} produced fewer such answers than not asking (2/36 against 4/36), the system turn buys one item in twenty-four ($p = 1.0$) — while two in-context examples move oracle accuracy by **+0.221** ($p = 9.3e{-}09$).

Consequence for this paper. A low exact-match score is not by itself evidence of a capability failure — over 1424 generations exact match recovers 0.038 of the answers where a last-number rule recovers 0.240 at half the false-positive rate. We therefore state E2's counting result as failure under exact-match scoring at length ≥ 8, not as a capability bound; the prediction it agrees with [4, 5] concerns state tracking, which E3 addresses without a scoring rule in the way.

§6. The Regime and the Answer Occupy Different Windows

The rotation statistic is computed over the tail, while the answer is best-ranked at a median unroll of about four. Measuring it on both windows separates two conflated facts. Over the decision window (unrolls 0–11) rotating and settling prompts are indistinguishable, 0.2292 against 0.2296, a separation of **0.0004**; over the tail they separate completely, 0.862 against 0.298, a separation of **0.565**.

A sliding 12-unroll window dates the divergence between window starts 12 and 16 (Figure 1). The early window is not quiet — its non-DC power exceeds the tail's by two orders of magnitude (1.6×10^5 against 1943 and 59) — it is energetic and simply carries no period-6 structure. The answer is therefore fixed roughly twelve unrolls before the regime is measurable.

Scope. This is a timing claim, not a causal one, and it bounds the instrument as much as the model: P_6 needs two cycles, so 12 unrolls is the earliest any period-6 statistic can speak. A behavioural null from manipulating the regime is correspondingly weak — a causally induced regime change altered the answer in 1 of 18 paired units, acting on a property that did not yet exist when the readout settled.

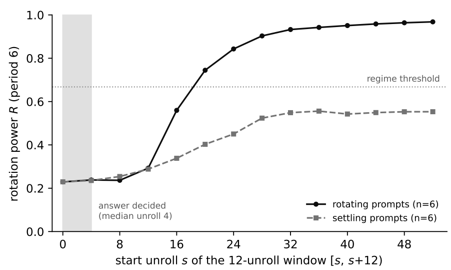

Figure 1. Sliding 12-unroll rotation power, median over six rotating and six settling prompts. The axis is the window start: each point summarises unrolls $[s,\ s{+}12)$. The families are indistinguishable until the window beginning at unroll 16, by which point the answer is long settled (shaded; median unroll 4). Higher is more period-6 power.

§7. Two-Scale Dynamics and Step Orthogonality

On synthetic trajectories at depths 2–5, mean acceleration grows $4.89 \to 7.45$ (+52%, $\rho = 0.997$ against depth) while step orthogonality stays near -0.5 and the two-hit exit fraction is fixed at 50%. As a depth indicator acceleration (0.997) far outperforms steps-to-settle (0.18) and winding (0.053).

On real FineWeb text the same measurement gives step orthogonality -0.6870 ± 0.0993 ($133.9°$), acceleration 1.8360 ± 0.0548, step norm 13.7350 ± 3.1820 and a two-hit exit at step 5 — oscillatory dynamics, successive steps anti-aligning, rather than the spiral ($\cos \approx 0.5$–0.65) of [3]. More useful, the measurement is confounded with step norm unless the step is normalised: before normalisation orthogonality is not stably achieved, after it the correlation with step norm falls to -0.0436. We take step normalisation as the robust finding and restrict the comparison with [3] to this FineWeb arm, the synthetic arm being generated by a construction that is not the model studied.

§8. Discussion

Convergence is by design, and the two rotation measures differ in scope. Randomised latent initialisation promotes convergence and every answer-token trajectory in E1–E5 contracts, consistent with [2]. This is not in tension with E6: every E1–E5 prompt uses one instruction style, and E6 shows the regime is selected by wording at fixed task, answer and token count — so "all trajectories settle" is what E6 predicts there, making E1–E5 a replication within the settling class, not evidence that the model never rotates. The instruments differ accordingly: winding is a 2D-PCA quantity fitted to the trajectory it judges and confounds damping with rotation, a weak basis for a negative; P_6 carries a held-out-validated threshold and can support one.

What H2 does and does not survive. The winding operationalisation — winding number growing with required reasoning steps — is refuted on these prompt sets (E1, E2, E5). H2 as the adaptive-computation literature means it is a different claim and is not testable here at all: per-position ponder time rises with difficulty only in models trained with a halting cost [7–9], whereas num_steps unrolls the whole sequence together, so every position receives exactly r and there is nothing to allocate. Our null is evidence against the operationalisation, not the hypothesis.

H3, jointly. E3 shows effective computation grows with state retention, not prompt length. Independently the running count is linearly decodable from the latents, an untrained model carrying the lagged count at cross-validated 0.7498 against the trained model's 0.7175 — the register is architectural, not learned. Jointly: the model spends more compute on state-retention tasks and the state survives contraction. What fails is H3's strong form, that contraction implies no running count; E3 is untouched. The contraction rate is also family-level, spanning 0.8239 to 0.9181, and should not be quoted as one constant.

H1: both, at different levels. Drift is excluded by construction, every state being a normalisation output on a sphere of radius 76.386 against which two random points lie 108.03 ± 0.75 apart. A period-4 cycle exists across the four core blocks; within a block the state settles or rotates by instruction token (last-step residual 2.9152 against 0.0185).

Reliability. Several results here were produced by instruments rather than by the model. One behavioural null was vacuous — its correctness variable was false in 432 of 432 orbits — and re-tested on an outcome that varies the conclusion returned. A geometry headline was withdrawn by its own registered norm control, and a spectral run refused by its own consistency gate. A linear probe of the class used for several of our nulls cannot recover a label that is a guaranteed deterministic function of its own inputs (0.690 against a 0.600 baseline, where the correct statistic reaches 0.980), so those nulls are uninformative rather than negative.

§9. Limitations

- Answer-token focus. E1–E5 examine the final token, while the authors report orbits on question and digit tokens; E7 checks this for the regime labels and finds no bias, but the E1–E5 correlations are not re-measured per position.
- Proxy metrics. 2D-PCA winding is sign-arbitrary and basis-dependent, and steps-to-settle conflates a small step with convergence; $\rho(J)$, the correct quantity for H3, is not computed.
- Small samples, single seeds; configuration. Some apparent signals collapsed under multi-seed testing, power is insufficient at the 0.02–2.8% rate of [2], and we did not use the authors' system prompt or $r = 128$ — though the system turn is at least not the missing ingredient ($p = 1.0$, above).

- Shared bias; cross-checkpoint comparison. Our benchmark numbers and the published figure share an uncorrected surface-form bias from likelihood scoring, and comparisons with two related works are cross-checkpoint or cross-protocol.
- Timing, not causation. §6 says when the properties become measurable, not whether one influences the other. A behavioural test of H3 by injecting a donor state at h_0 failed three times and is not reported.

Next steps. Compute $\rho(J)$ by power iteration with Jacobian-vector products [10] and reclassify regimes with it; extend hooks to all positions and reproduce the authors' configuration at $r = 128$; replace per-trajectory 2D winding with a global projection; probe query–key alignment [11]; and apply the steering vector at all positions, the unrun discriminator for its null.

§10. Conclusion

Answer-token trajectories converge, and their effective computation grows with state retention rather than prompt length. The conspicuous geometric structure is nevertheless selected by the instruction's surface form rather than the task, and controlled from one embedding row. Much of what we and the standard protocols scored as capability was the model's willingness to emit its answer where a scorer looks. Whether the regime affects behaviour remains open; the experiment settling it must act earlier in the trajectory than ours did.

§11. Contribution

- Alexander Shiianov — extraction infrastructure and hooks; winding, steps-to-settle, shape classification, persistent homology; experiments E1–E5 including the length-matched control and phase map; correlation analysis.
- David Sverdlov — two-scale metrics; the FineWeb orthogonality reproduction and the step-normalisation finding; acceleration as a depth indicator.
- Arsenii Varaksin — rotation power and its validation; regime experiments E6–E7 with the embedding-row and parameter-swap interventions; the format-versus-capability decomposition E8 and ARC

protocol arms; the window split; the steering null; the instrument-null and retraction record.

- Serguei Barannikov (mentor) — problem formulation and hypotheses H1–H3; N. Kurdyukov (mentor) — supervision.

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Траектории рассуждений в трансформере с рекуррентной глубиной

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Аннотация

Исследуются латентные траектории Huginn-3.5B. Эффективное число шагов вычисления растёт с необходимостью удерживать состояние, а не с длиной запроса; одно слово в инструкции переключает отображение между сходимостью и затухающим вращением с периодом 6; а значительная часть измерений качества фактически измеряла формат ответа, а не его наличие.

Ключевые слова: трансформеры с рекуррентной глубиной, латентные рассуждения, интерпретируемость, протоколы оценки.